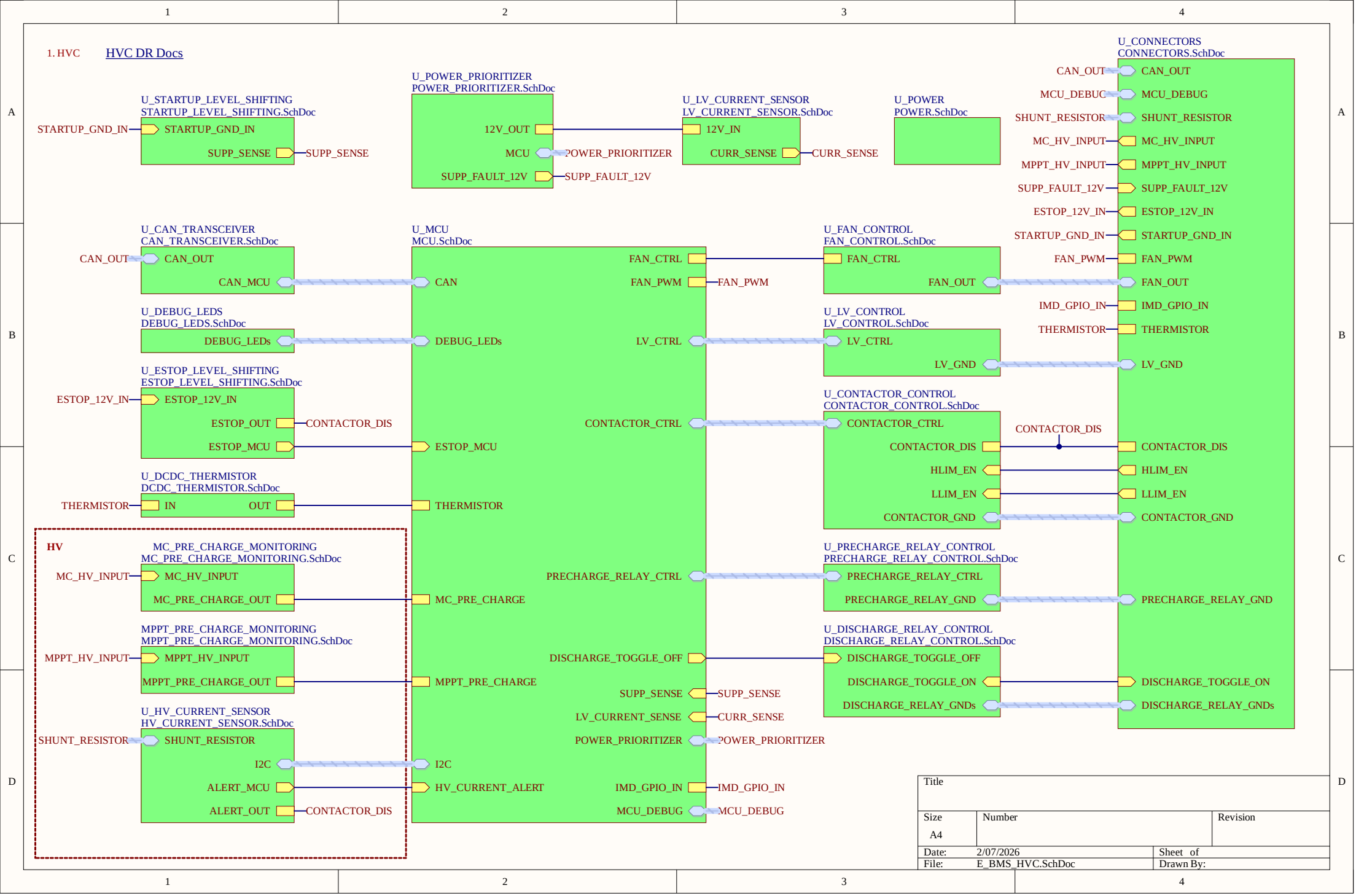
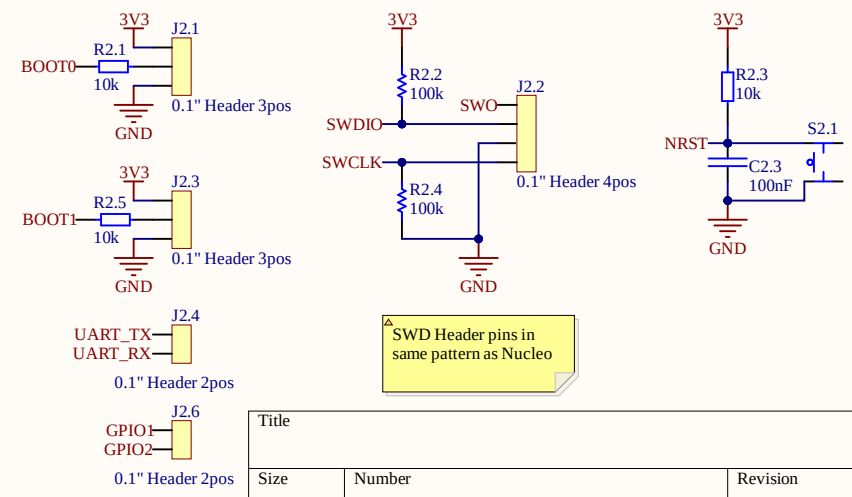
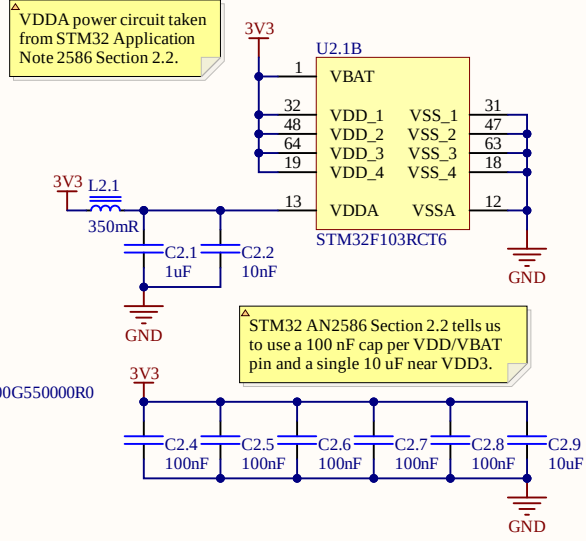
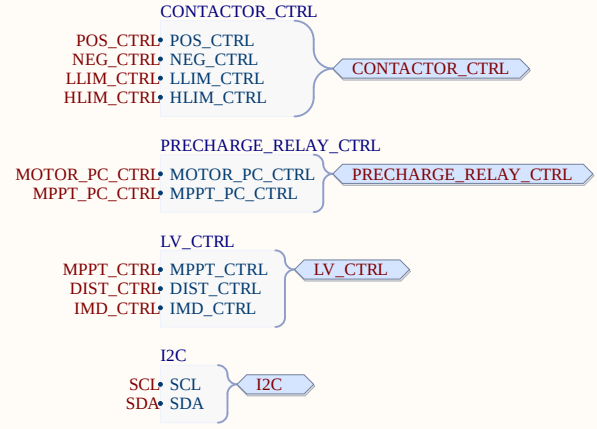
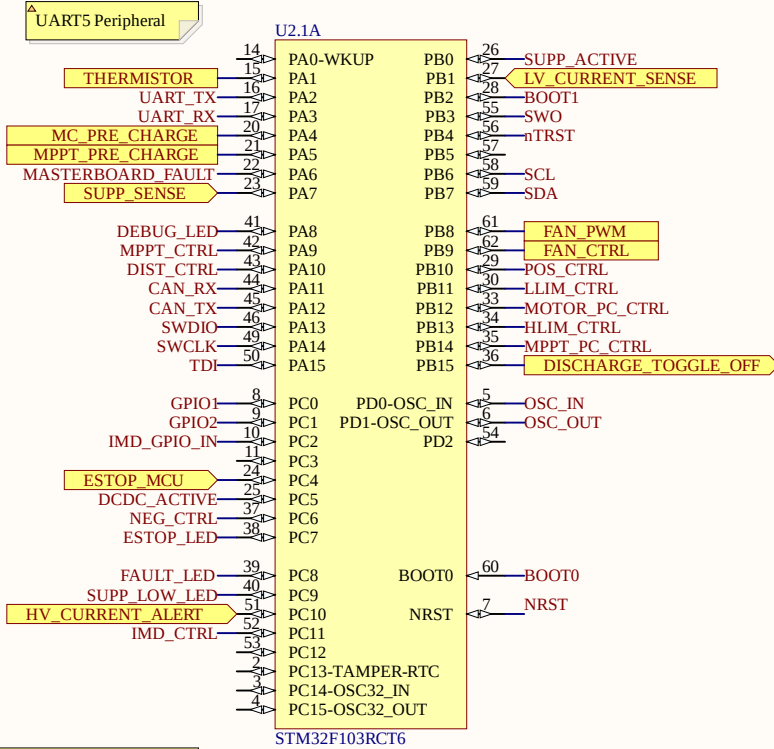
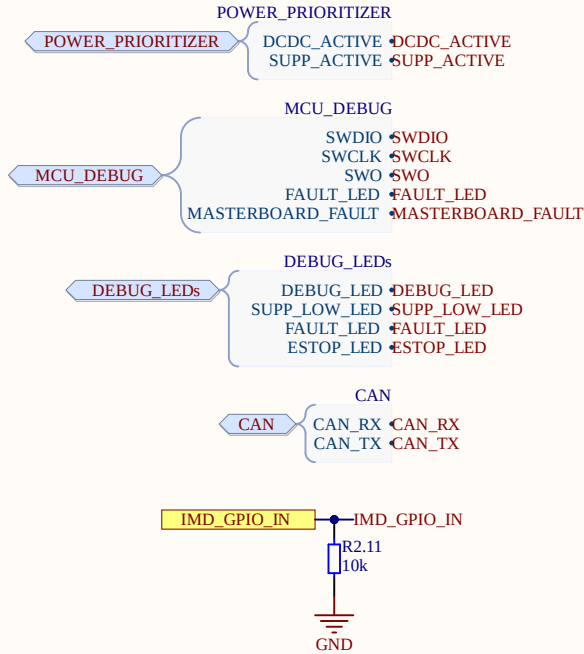




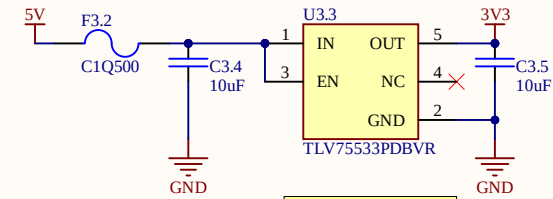
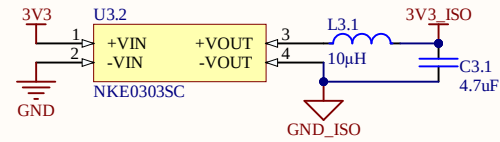
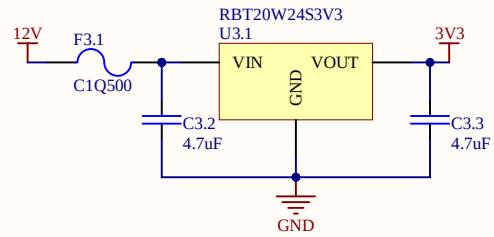
2. MCU



Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	MCU.SchDoc	Drawn By:

3. POWER

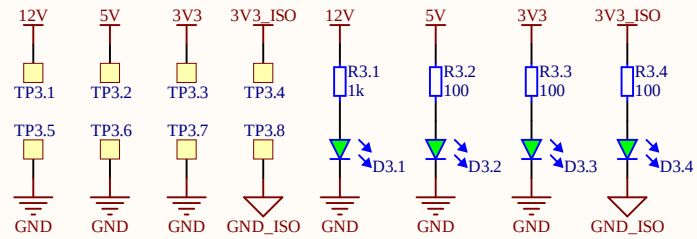
[Monday Update Link](#)



Inductor and Capacitor values specified in datasheet:
<https://www.murata.com/products/productdata/8807030915102/kdc-nke.pdf>

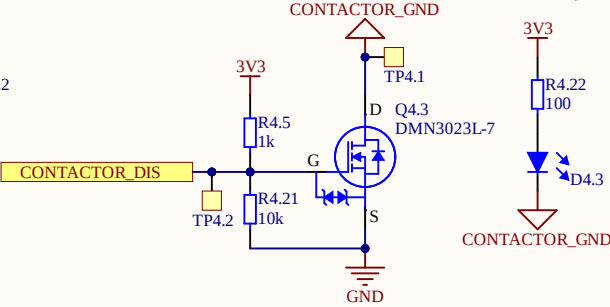
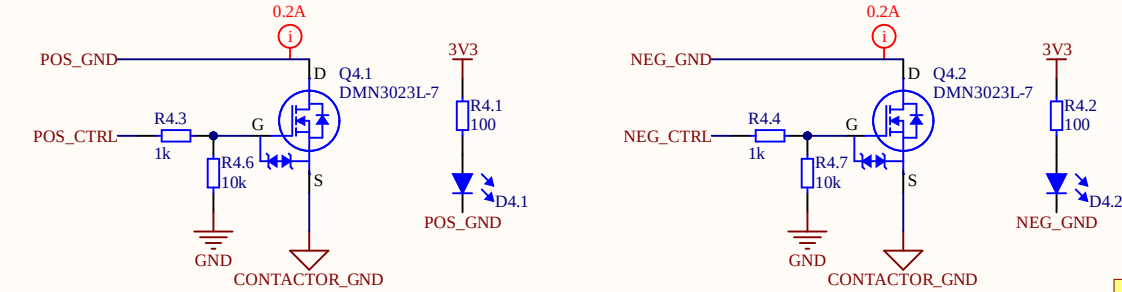
Datasheet Link

△ J-Links supply us with 5V, we need an LDO to get 3V3.



Title		
Size A4	Number	Revision
Date:	2/07/2026	Sheet of
File:	POWER.SchDoc	Drawn By:

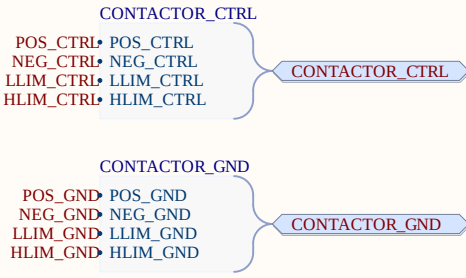
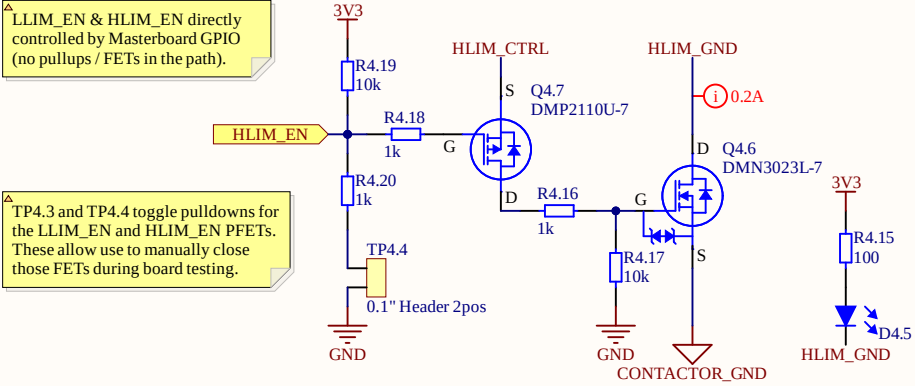
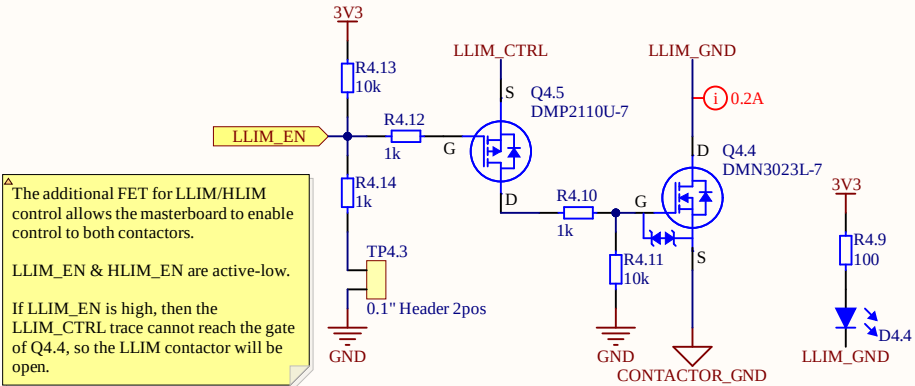
4. CONTACTOR CONTROL



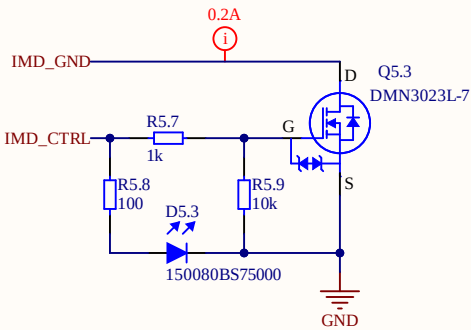
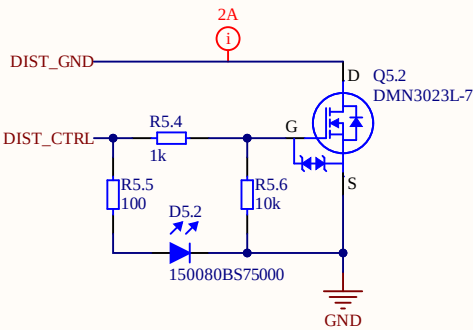
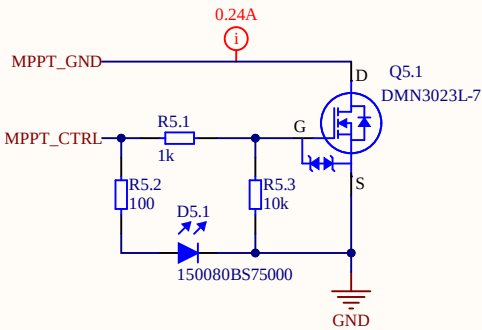
CONTACTOR_EN TRUTH TABLE

STATE	CONTACTOR EN
Nominal	Floating
Fault	0 V

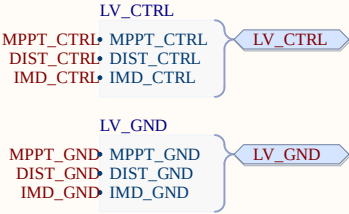
[Monday Update Link](#)



Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	CONTACTOR_CONTROL.SchDoc	Drawn By:

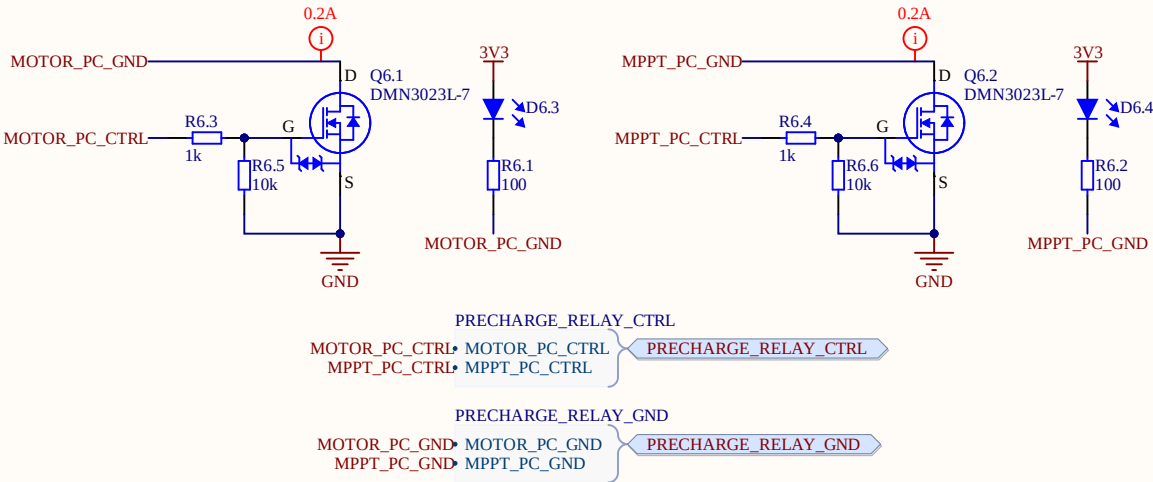


Use of a 1k current limiting resistor then a 10k pulldown is important. If it's 1k 1k, you get a 50% voltage divider and NMOS gate voltage may not be driven about Vgs(th).



Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	LV_CONTROL.SchDoc	Drawn By:

6. PRECHARGE RELAY CONTROL



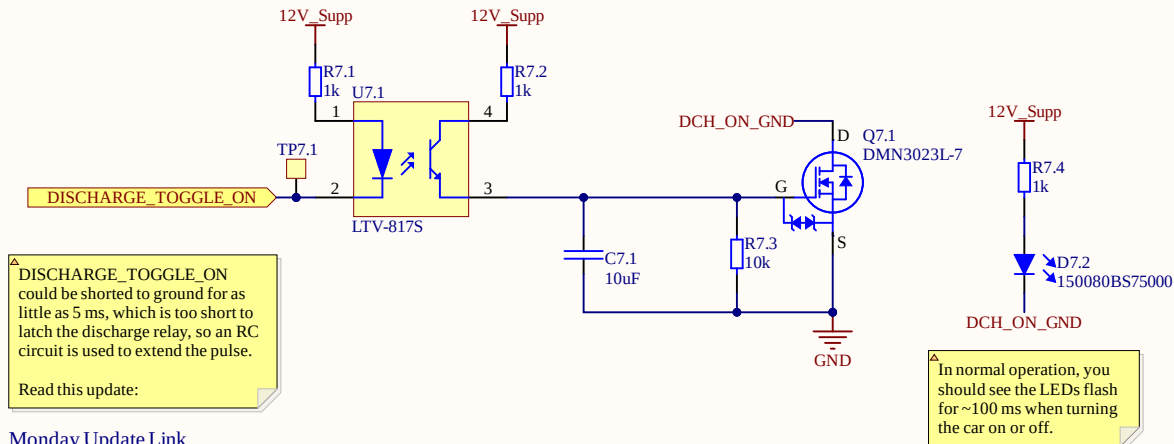
Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	PRECHARGE_RELAY_CONTROL.SchDoc	Drawn By:

7. DISCHARGE RELAY CONTROL

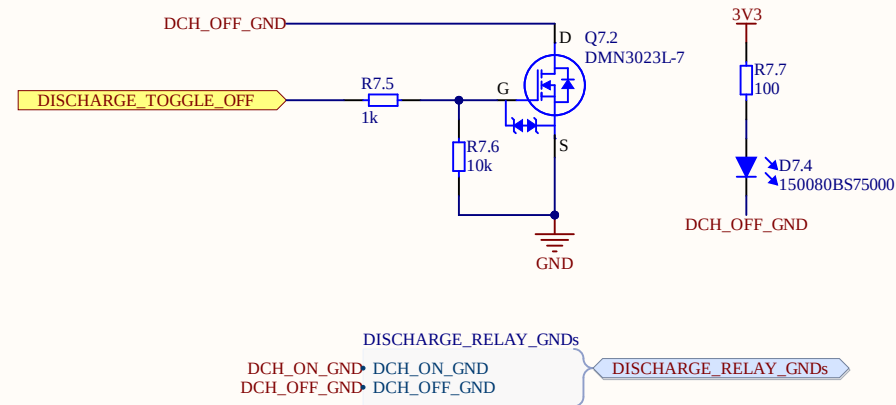
[Monday Update Link](#)

A

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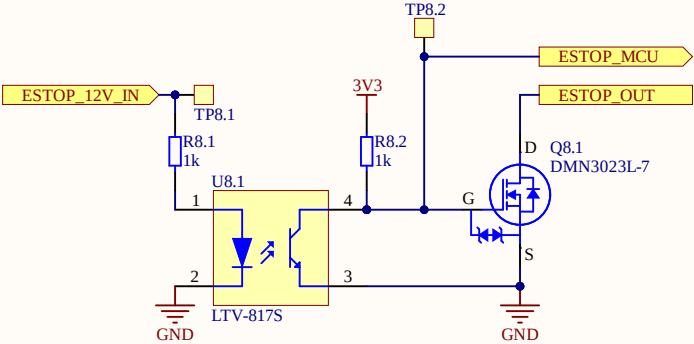


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Title		
Size A4	Number	Revision
Date:	2/07/2026	Sheet of
File:	DISCHARGE RELAY CONTROL.Sch Drawn By:	



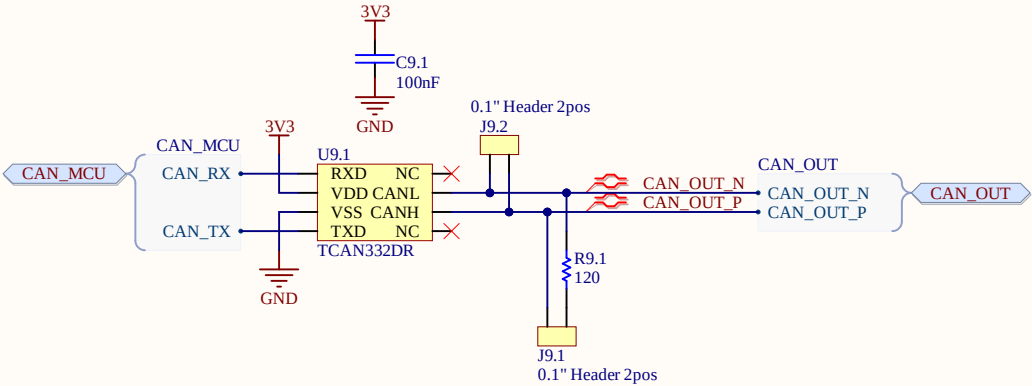
ESTOP Truth Table

STATE	ESTOP 12V IN	ESTOP MCU	ESTOP OUT
Nominal	12 V	0 V	Floating
Fault	0 V	3.3 V	0 V

ESTOP_OUT goes to CONTACTOR_DISABLE, when it's pulled to GND the contactors cannot be closed.

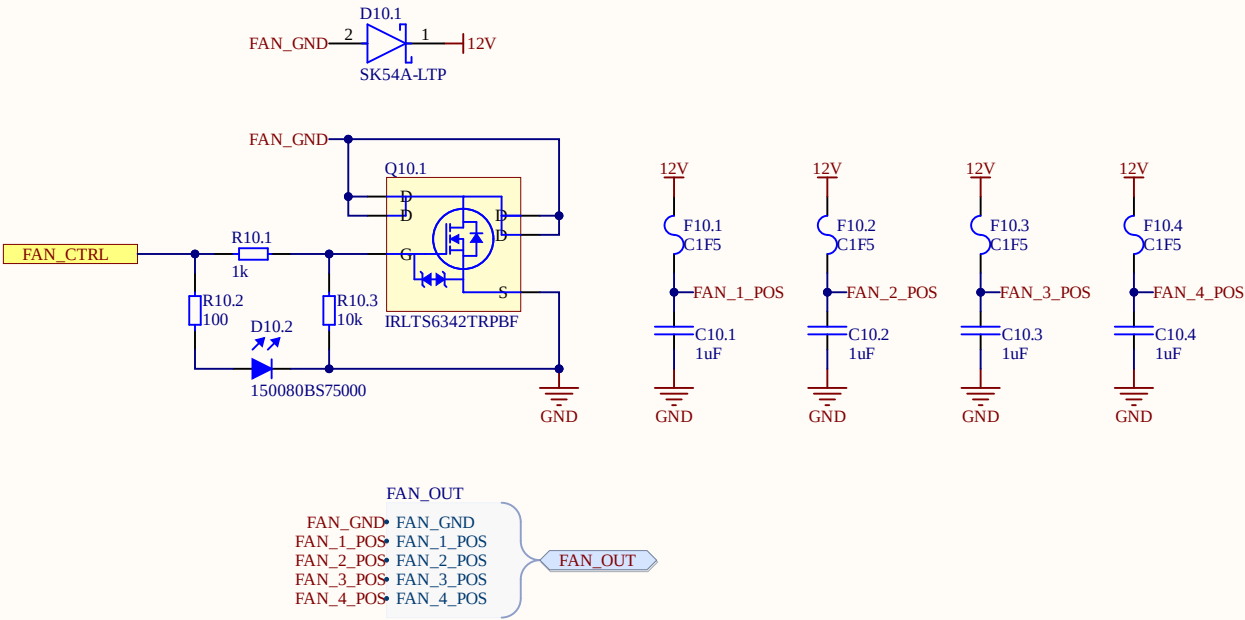
Title		
Size A4	Number	Revision
Date:	2/07/2026	Sheet of
File:	ESTOP_LEVEL_SHIFTING.SchDoc	Drawn By:

9. CAN TRANSCEIVER



Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	CAN_TRANSCEIVER.SchDoc	Drawn By:

⚠ The SK54A Schottky Diode is rated for 5A, and best practice is to size flybacks for max load current, so fans should not exceed 5 A.



⚠ The decoupling capacitors are required for fan PWM, and must go to GND not FAN_GND because FAN_GND is floating when the NMOS is open, meaning the capacitor voltage would be undefined if it went to FAN_GND.

Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	FAN_CONTROL.SchDoc	Drawn By:

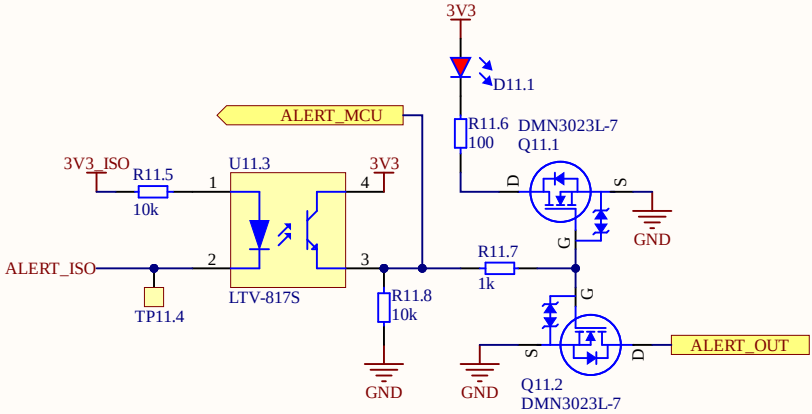
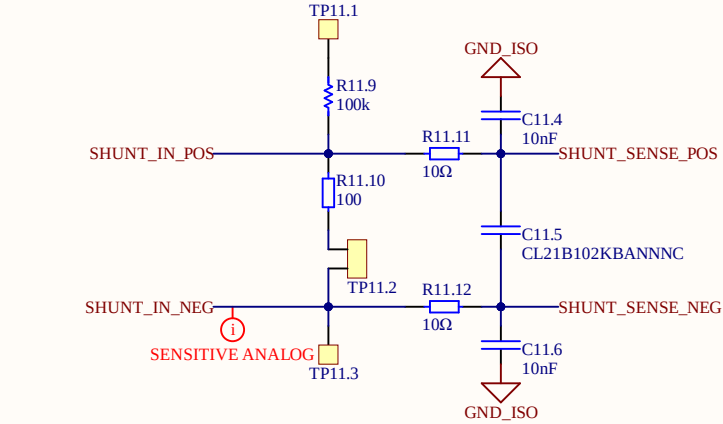
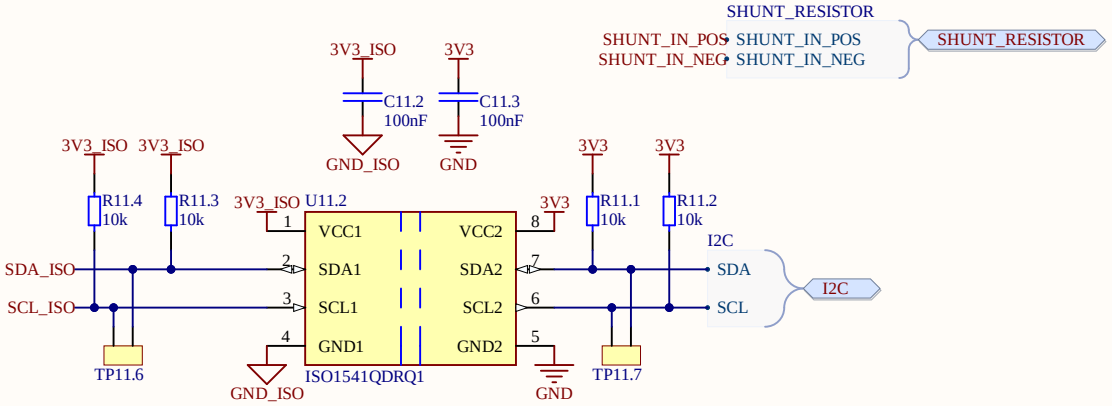
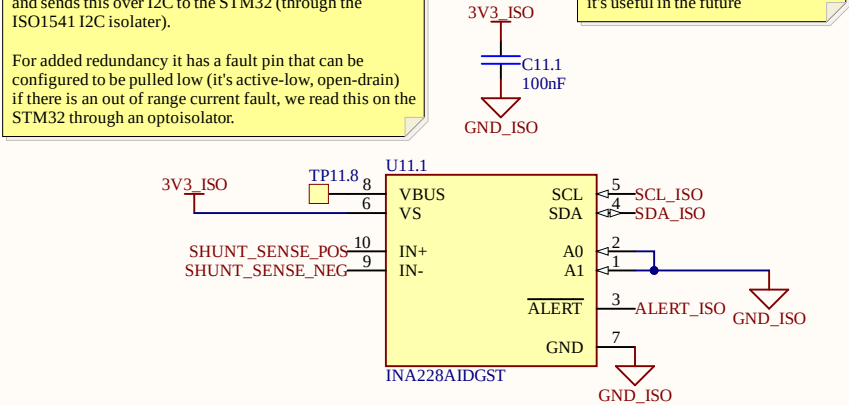
11. CURRENT SENSING

The INA228 is a current sense amplifier with an internal differential ADC to measure voltage over the shunt resistor. It internally translates this to a current output, and sends this over I2C to the STM32 (through the ISO1541 I2C isolator).

For added redundancy it has a fault pin that can be configured to be pulled low (it's active-low, open-drain) if there is an out of range current fault, we read this on the STM32 through an optoisolator.

VBUS_SENSE pin on the INA228 can sense a 0-80 V input with ~200 uV precision. Breaking this out if it's useful in the future

[Schematic Design Monday Update Link](#)
[Shunt Resistor + Component Selection Monday Update\(s\) Link](#)



100k + 100 ohm voltage divider used for scrutineering.

PSU should be connected over both test points.

Expected shunt voltage:
 $V = IR$. 60 A * 100 uohms = 6 mV

Worst case scenario: $V_{in} = 30$ V, $V_{out} = 30$ mV. The INA228 can sense +/- 163.84 mV, so even with the max power supply voltage we're not outside the ADC range.

Shunt Sense RC Filter Values:

$C_{diff} = 1/(2\pi * 2R_{IN} * f_{cutoff})$
 $C_{CM} = 1/(2\pi * R_{IN} * f_{cutoff})$

$f_{cutoff_diff} = 10$ kHz $f_{cutoff_CM} = 1$ MHz
 $R_{IN} = 10$ ohm

These values result in:
 $C_{diff} = 0.796$ nF $C_{CM} = 15.9$ nF

Read the Monday Update:

[Filtering Monday Update Link](#)

ALERTs Truth Table

STATE	ALERT ISO	ALERT MCU	ALERT OUT
Nominal	Floating	0 V	Floating
Fault	0 V	3.3 V	0 V

ALERT_OUT goes to CONTACTOR_DISABLE, when it's pulled to GND the contactors cannot be closed.

When ALERT_ISO goes low (it's open-drain active-low), ALERT_OUT is pulled to GND.

Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	HV_CURRENT_SENSOR.SchDoc	Drawn By:

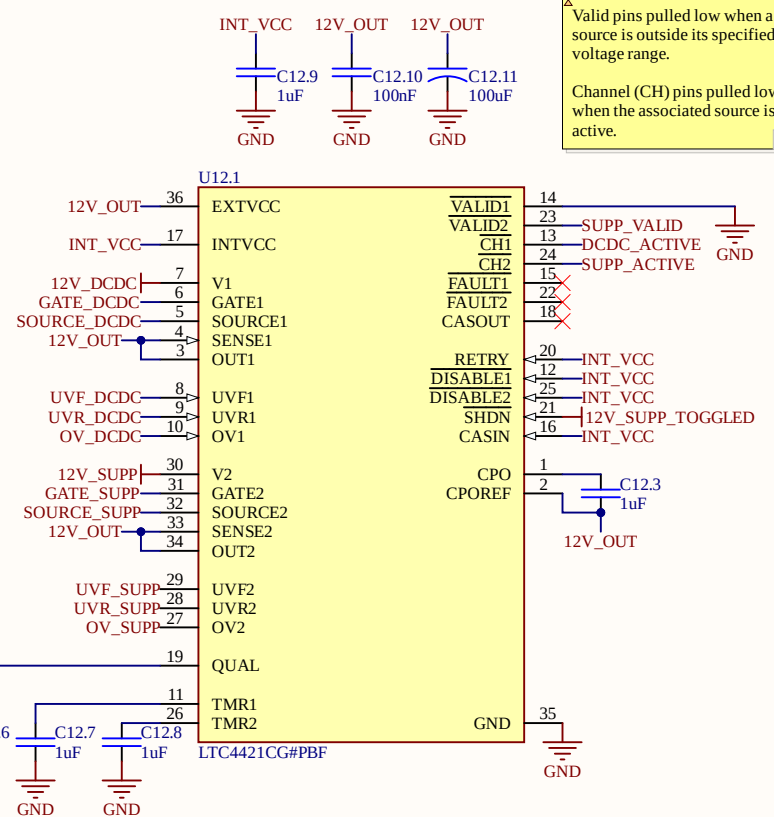
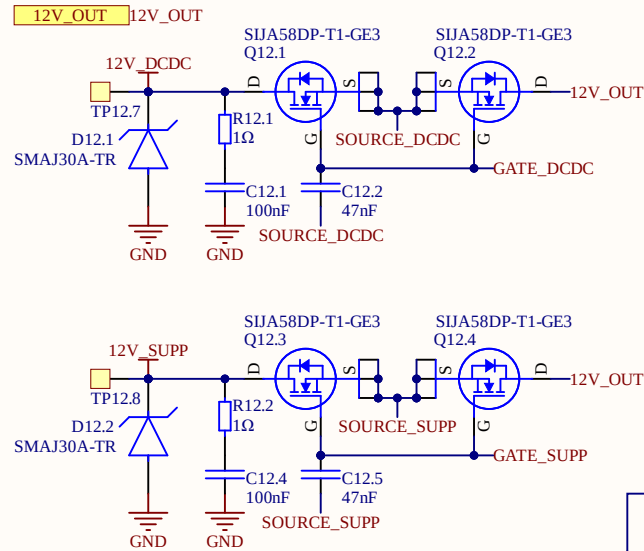
12. POWER PRIORITIZER

RC Snubber Math Here:
<https://ubcsolar26.monday.com/boards/9702086049/pulses/18080990623/posts/4800401935>

Back-to-back NMOS's prevent current flow in both directions when gate is low, taking into account the NMOS's body diode.

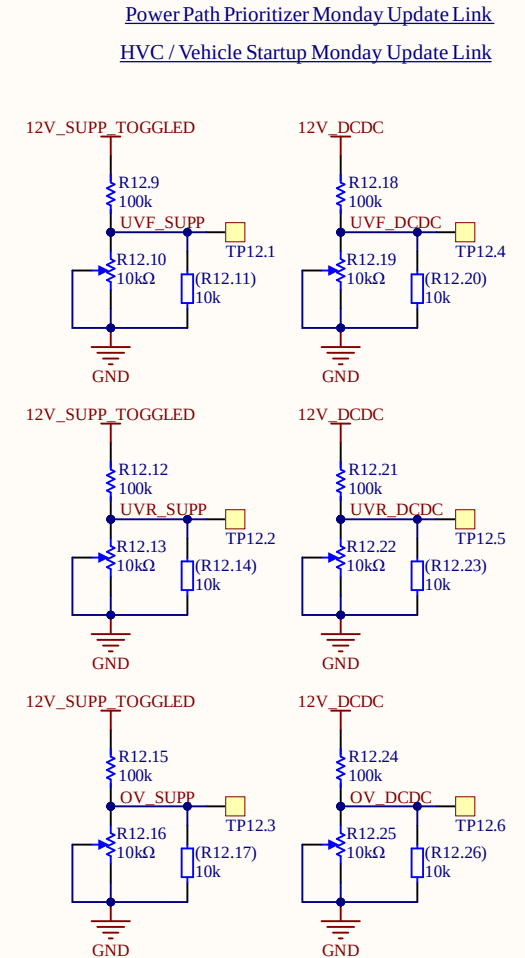
RC Snubber Circuit Math Here:

[Snubber Monday Update Link](#)



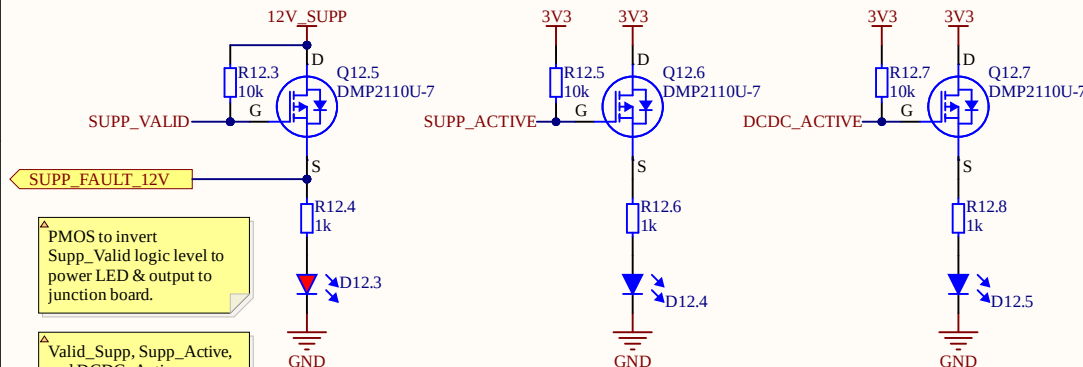
Valid pins pulled low when a source is outside its specified voltage range.

Channel (CH) pins pulled low when the associated source is active.



[Power Path Prioritizer Monday Update Link](#)

[HVC / Vehicle Startup Monday Update Link](#)



PMOS to invert
Supp_Valid logic level to
power LED & output to
junction board.

Valid_Supp, Supp_Active, and DCDC_Active are open-drain and active low.

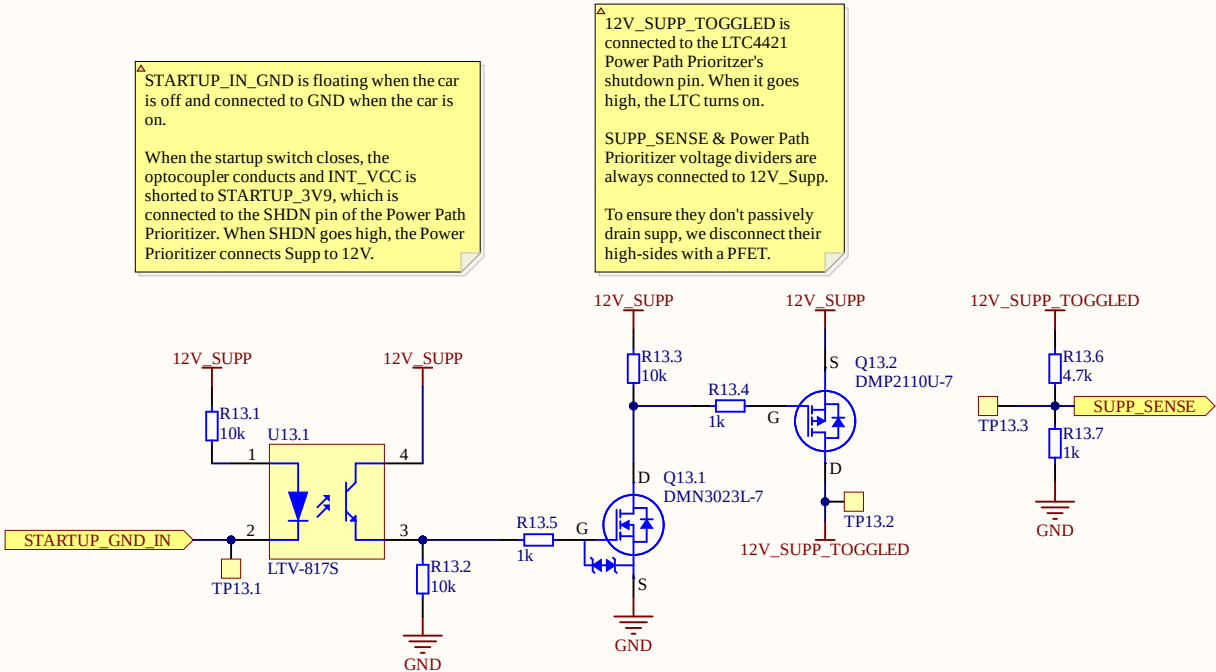
△ DNP's for if we ever don't trust the trimmer pots or future BMS members to not mess with the trimmer pots. Ie. set voltage thresholds in hardware.

Undervoltage falling/rising (for hysteresis, see the Monday updates linked below), and Overvoltage thresholds set with voltage dividers below.

When a voltage is $>0.5\text{ V}$, the corresponding fault (UVF, UVR, OV) is triggered in the LTC4421.

Title		
Size A4	Number	Revision
Date:	2/07/2026	Sheet of
File:	POWER_PRIORITIZER.SchDoc	Drawn By:

13. STARTUP LEVEL SHIFTING



STARTUP_IN_GND is floating when the car is off and connected to GND when the car is on.

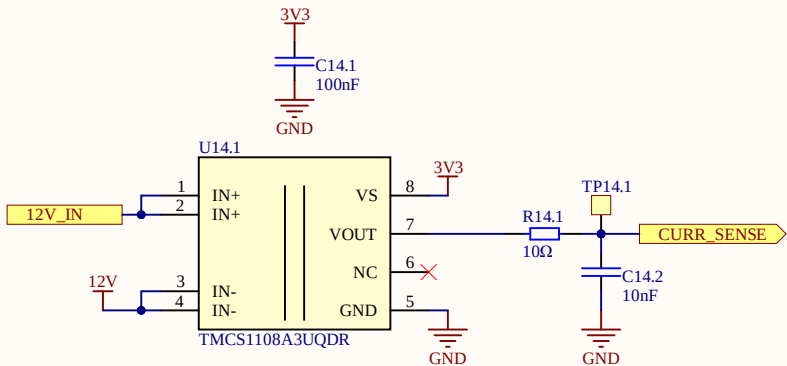
When the startup switch closes, the optocoupler conducts and INT_VCC is shorted to STARTUP_3V9, which is connected to the SHDN pin of the Power Path Prioritizer. When SHDN goes high, the Power Prioritizer connects Supp to 12V.

12V_SUPP_TOGGLED is connected to the LTC4421 Power Path Prioritizer's shutdown pin. When it goes high, the LTC turns on.

SUPP_SENSE & Power Path Prioritizer voltage dividers are always connected to 12V_Supp.

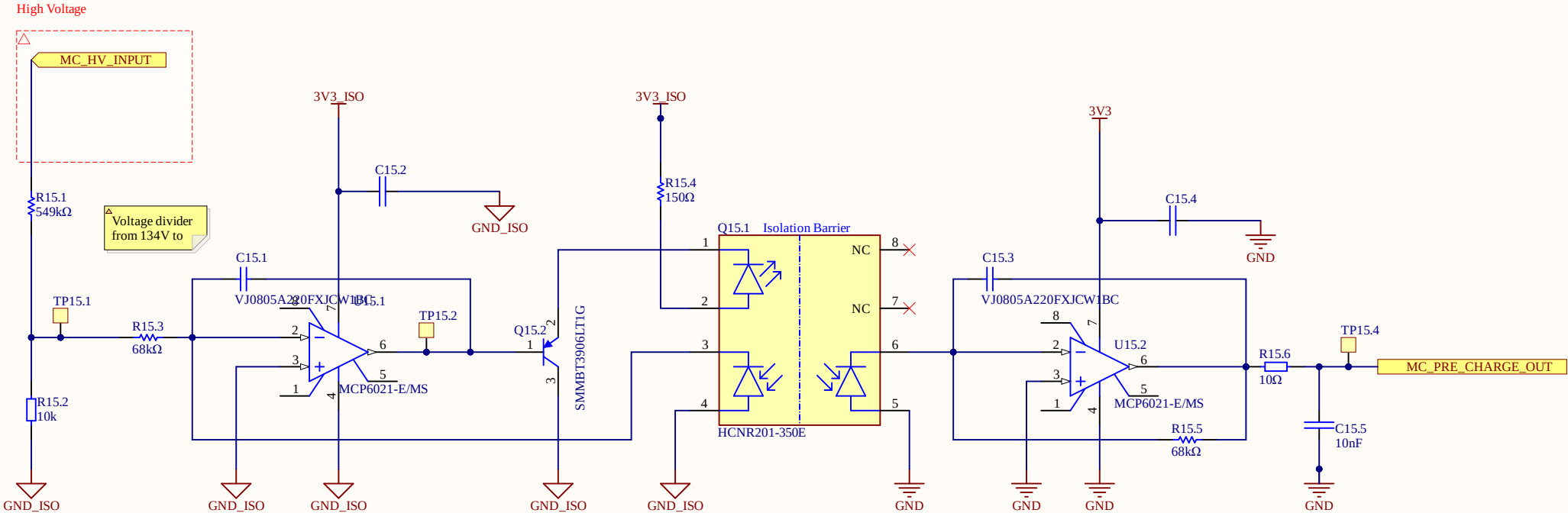
To ensure they don't passively drain supp, we disconnect their high-sides with a PFET.

Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	STARTUP_LEVEL_SHIFTING.SchDoc	Drawn By:

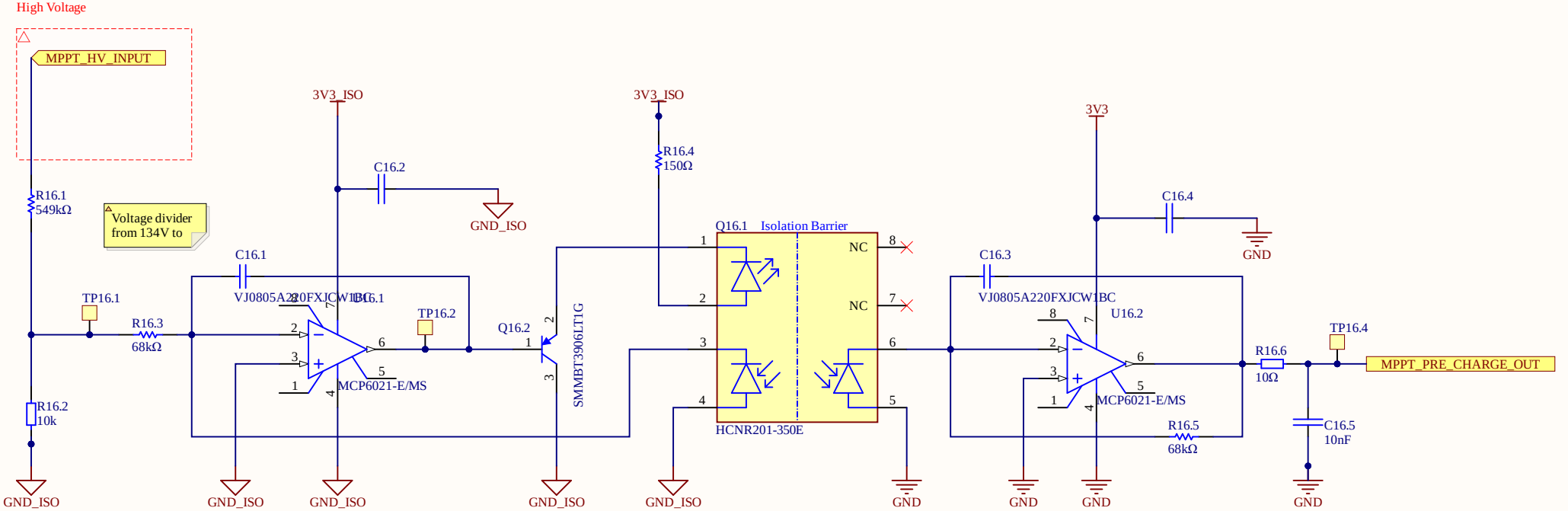


Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	LV_CURRENT_SENSOR.SchDoc	Drawn By:

15 PRE_CHARGE_MONITORING

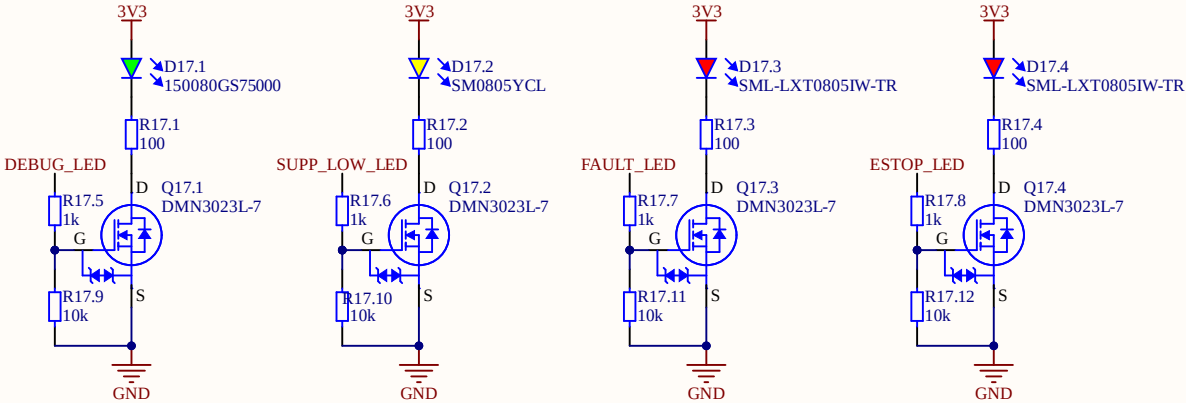


Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	MC_PRE_CHARGE_MONITORING.Sch	Drawn By:



Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	MPPT_PRE_CHARGE_MONITORING.SchDoc	Drawn By:

17.DEBUG LEDs



My general opinion on LED colours:

- Green = Debug (eg. Debug LED or 3V3)
- Blue = Control (eg. LLM contactor closed)
- Yellow = Warning (eg. supp low)
- Red = Fault (eg. over current)

DEBUG_LEDs

- DEBUG_LED
- SUPP_LOW_LED
- FAULT_LED
- ESTOP_LED

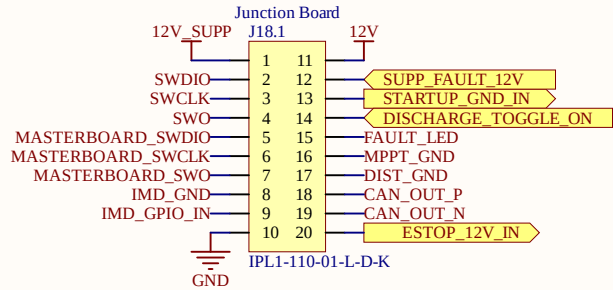
DEBUG_LEDs

Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	DEBUG_LEDS.SchDoc	Drawn By:

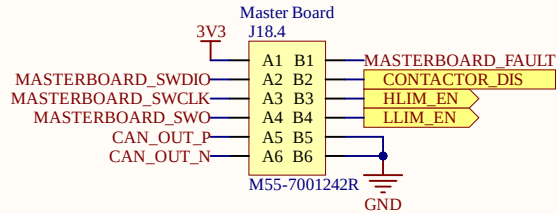
18. CONNECTORS

[Monday Update\(s\) Link](#)

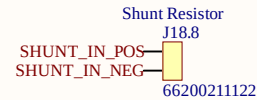
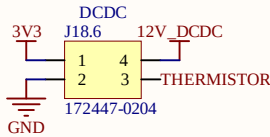
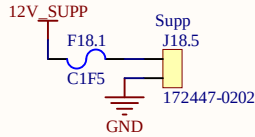
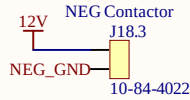
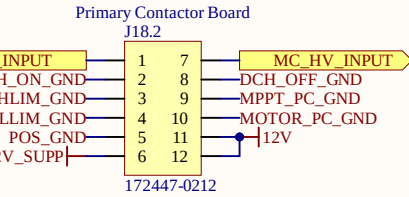
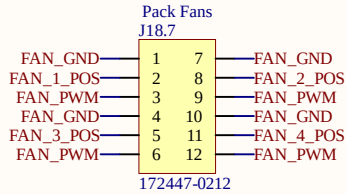
A



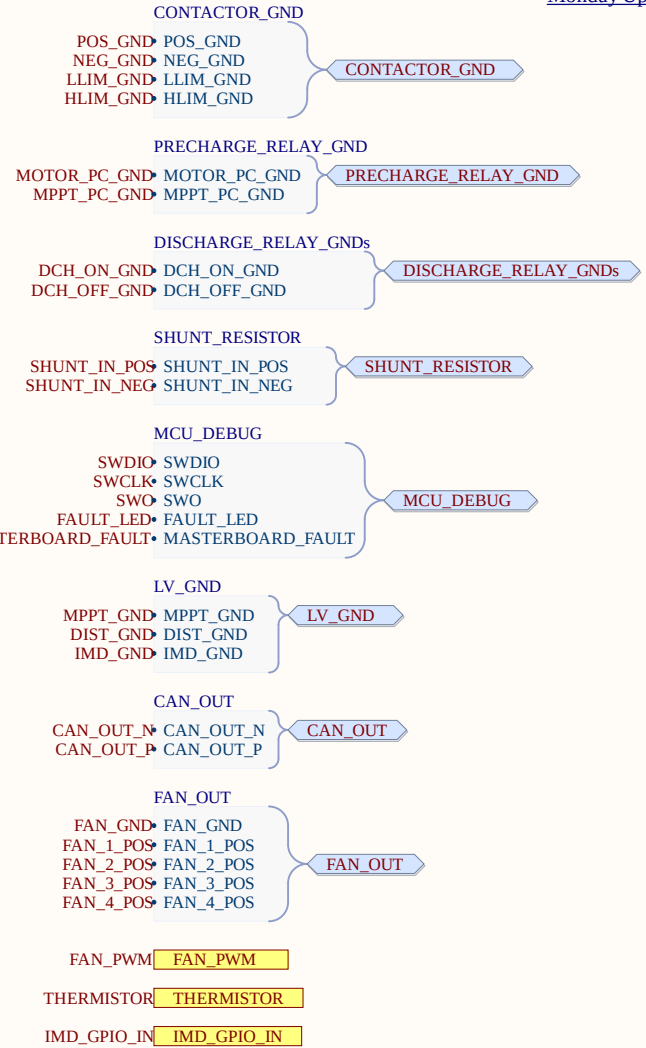
B



C



Usually 12V_Supp is supplied from the Junction Board connector, but when benchtop testing with the JCT the additional 2 pin Supp connector can be used. Hence, it needs its own fuse.



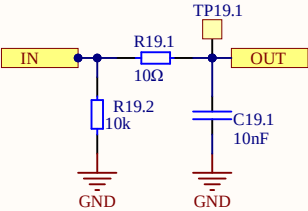
A

B

C

D

Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	CONNECTORS.SchDoc	Drawn By:



Title		
Size	Number	Revision
A4		
Date:	2/07/2026	Sheet of
File:	DCDC_THERMISTOR.SchDoc	Drawn By: